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RESEARCH ARTICLE

# Optimization of Customer Feedback Summarization Using Large Language Models (LLM) and Advanced Retrieval-Augmented Generation

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**ABSTRACT** Customer feedback, often shared through online reviews, plays a crucial role in shaping business strategies. However, the overwhelming volume of such reviews poses two major challenges: valuable insights often go unnoticed, and manual analysis introduces human bias. To address this, we propose a system that leverages large language models (LLMs) integrated with the LangChain framework to answer natural language queries over customer reviews. A synthetic dataset was created to resemble food delivery reviews typically seen on the Play Store and was stored in a vector database. On receiving a user query, relevant reviews are retrieved using advanced Retrieval-Augmented Generation (RAG) techniques, namely Hierarchical Chunk Retrieval and RAG Fusion that are further refined using the Declarative Self-improving Python (DSPy) framework to generate accurate, grounded responses. The system was evaluated using LLaMA-3-8B-InstructLite, GPT-3.5-Turbo, and Gemini-1.5-Pro, and compared against two non-LLM baselines: BM25 and a fine-tuned BERT model. Results show that our LLM-based pipeline outperforms by an average 15% in semantic and factual accuracy. Component-level analysis showed that enhanced retrieval strategies showed aggregate improvements of 4.9% in semantic relevance, 12.1% in lexical coverage, and 9.9% in factual consistency over traditional RAG. Further integration of DSPy led to an additional 10.8% boost in linguistic fluency and a 9.0% gain in factual alignment. Among the evaluated models, Gemini-1.5-Pro combined with RAG Fusion and DSPy produced the most fluent and factually accurate responses, demonstrating the effectiveness of combining hybrid retrieval with LLM-driven reasoning for query-based feedback response system.

**INDEX TERMS** Customer reviews, GPT-3.5-Turbo, LangChain, LLM, Llama-3-8B-Instruct-Lite, Gemini-1.5-Pro, RAG, DSPy, BM25, BERT, RAG fusion, hierarchical chunk retrieval.

## I. INTRODUCTION

In this digital era, customers frequently share their experiences, opinions, and concerns about products and services through online reviews. These reviews serve as valuable feedback for businesses, influencing product development, marketing strategies, and customer engagement. However, the exponential growth of user generated content has made it increasingly difficult for companies to process and extract meaningful insights from vast amounts of unstructured text

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data. Traditional methods of analyzing customer feedback, such as manual review and rule-based text processing, are not only time-consuming but also prone to inconsistencies and subjective interpretations. Additionally, human bias in feedback analysis can lead to skewed insights, potentially affecting business decisions. As a result, there is a growing need for automated, scalable, and unbiased techniques to effectively summarize and interpret customer opinions. To tackle the challenge of analyzing large volumes of customer feedback, various computational techniques have been explored. Conventional methods for analysing customer feedback primarily rely on classification or extraction techniques that offer limited, non-interactive insights. For instance, basic sentiment analysis typically uses machine learning models like Logistic Regression [1], Support Vector Machines (SVM) [2], or Random Forests [3] trained on labeled datasets to classify reviews as positive, negative, or neutral. While this is useful for gauging overall sentiment trends, it does not provide specific answers to targeted queries or insights about particular aspects of a product or service. Keyword extraction approaches like TF-IDF [4], [5] and Named Entity Recognition (NER) [6], [7] identify frequent or significant terms within the text but offer no reasoning, summarization, or query response capabilities.

To move toward question-answering capabilities, we implemented two representative non-generative response generation baselines. The first is BM25 [8], [9], a sparse lexical retrieval algorithm that ranks documents based on keyword overlap and inverse document frequency. Although efficient, it lacks semantic understanding and fails to capture context beyond word matching. The second baseline uses a fine-tuned BERT [10], [11], [12] model trained for extractive question answering. It processes the full review text directly and selects the most relevant span as the answer. While it captures local context well, it lacks retrieval capabilities and cannot scale to larger document collections or handle queries that require aggregating information across multiple reviews.

These limitations demonstrate the need for more advanced and flexible query-based feedback response systems. Our proposed solution integrates Large Language Models (LLMs) with hybrid retrieval techniques and multi-hop reasoning via the DSPy framework. LLMs enable deeper semantic understanding of complex queries, while Retrieval-Augmented Generation (RAG) [13], [14], [15] ensures that generated responses are grounded in the retrieved customer feedback. The DSPy-based multi-hop refinement further enhances answer quality by iteratively reformulating queries and incorporating additional context. This combined approach allows our system to deliver fluent, faithful, and contextually grounded answers to natural language questions over customer review—capabilities not supported by traditional classification, extraction, or span-based feedback response generation methods.

Key contributions of this work include:

- We extend the standard RAG framework by introducing a modified chunking strategy that groups semantically related review segments under parent identifiers. While based on existing RAG principles, this adaptation improves context coherence during retrieval and enhances the relevance of generated responses.
- We adopt and apply the RAG Fusion technique to customer feedback tasks, leveraging multiple reformulations of a query to retrieve and aggregate diverse evidence. Though derived from prior work, its application and tuning in this domain yield measurable performance gains.
- We integrate the DSPy (Declarative Self-improving Python) framework independently with both retrieval strategies to enable iterative query rewriting and multi-hop reasoning. This allows the system to refine responses based on dynamic query adjustments, enhancing factuality and depth.
- We evaluate the system using BERT Score, ROUGE, and RAGAS, capturing both linguistic quality and retrieval-grounded performance metrics.
- We benchmark the proposed methods against non-generative baselines using BM25 and a fine-tuned BERT model, demonstrating approximately a 15% improvement in answer quality and retrieval alignment using our LLM-based approaches.

The rest of the paper is structured as follows: Section II presents the literature survey. Section III outlines the implementation methodology. Section IV discusses the results and analysis. Section V highlights potential future work, and Section VI concludes the paper.

## II. LITERATURE SURVEY

Hua et al. [16] presents a novel approach integrating large language models (LLMs) with dynamic knowledge evolution for adaptive human-robot collaboration in complex product assembly. It emphasizes transforming robots into active cognitive collaborators, enhancing situational awareness, reasoning, and problem-solving in dynamic environments. Xiong et al. [17] highlight a multi-hop dense retrieval framework that improves accuracy and efficiency in complex question answering without relying on structured corpora. By comparing previous retrieval methods, they demonstrated better handling of complex multi-hop queries. RAG techniques further enhanced response quality by integrating recent information and reducing hallucinations. Wang et al. [18] discusses the evolution of RAG, highlighting advancements like modular designs, fine-tuning strategies, and multimodal retrieval to enhance efficiency and accuracy in specialized domains. "Hybrid with HyDE" is suggested to be implemented to achieve high accuracy. Arslan et al. [19] in their study focus on QA tasks in NLP. Afzal et al. [20] focus on four optimizations: child parent retriever, ensemble retriever, in context learning and multi-query using Llama2, Mistral, GPT-3.5 and GPT-4 models. Li et al. [21] highlights advancements in multi-query techniques, novel evaluation metrics, and the need for further optimization to improve faithfulness and coherence in generated content. Results indicate that long context is better than RAG for QA tasks. Chen et al. [22] introduces the review-then-refine framework, enhancing multi-hop question answering by dynamically decomposing sub-queries. It improves efficiency, reduces hallucinations, and sets a new standard for temporal multi-hop QA in RAG systems. Shi et al. [23] not only highlights the limitations of RAG but also introduced the GenGround Framework to solve a multi-hop question which proved to be superior. Khattab et al. [24] in their study state that RAG and Chain OF Thought helps improve the DSPy answer EM compared to the Vanilla few shot prompting. Asai et al. [25] in their study explores the advancements in general purpose retrieval augmented language models. The results state that general purpose retrieval augmented language models are more adaptable. Rackauckas et al. [26] in their study states that RAGElo derived from Elo ranking system which provides a CLI and python library which uses LLM's to evaluate retrieval results and responses from rag pipeline. Saha et al. [27] state that BERTScore and the RAGAS framework offer detailed quantitative analysis of response relevance and semantic accuracy. Their QuIM-RAG system, built using a custom dataset and LLaMA3 8B model, shows improved accuracy and relevance. The literature survey explores advancements in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) systems across various domains. Key areas of focus include optimizing prompt engineering, integrating dynamic knowledge for adaptive human-robot collaboration, and enhancing accuracy in complex question-answering tasks. Several studies emphasize multi-hop retrieval frameworks, improved retrieval techniques, and strategies like review-then-refine, modular designs, and hybrid methods for higher efficiency and reduced hallucinations. Interdisciplinary collaboration, standardized evaluations, and further research in fine-tuning and multi-query techniques are highlighted to improve LLMs' performance and safety. Tools like DSPy and frameworks like GenGround aim to address limitations, offering promising solutions for better AI system integration and response relevance.

## III. IMPLEMENTATION METHODOLOGY

The complete methodology and its architectural flow are depicted in Fig. 1.

### A. DATA PREPARATION AND PREPROCESSING

The dataset used in this study comprises approximately 100,000 synthetic customer reviews created to resemble real-world feedback found on platforms like the Play Store. These reviews were generated to reflect natural language, sentiment variety, and structure commonly observed in food delivery app reviews. The average review length is about 75 words, with an estimated sentiment distribution of 40% negative, 30% neutral, and 30% positive. A few representative examples are provided at: sample dataset link. To avoid potential legal or ethical concerns, no real-time user data was scraped or used; all content was synthetically created without any identifiable user information or brand references. The dataset was preprocessed by segmenting the textual

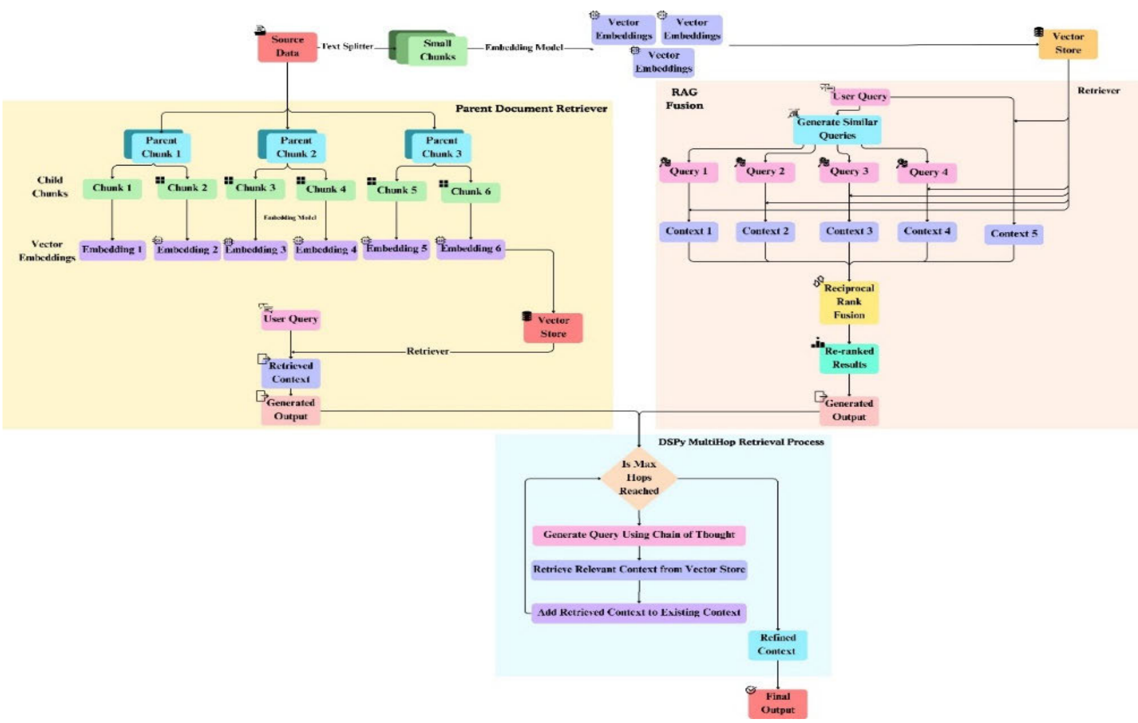


FIGURE 1. Architectural diagram representing flow of execution.

content into smaller, coherent chunks using the recursive text splitter technique. These chunks were then embedded into a vector representation using a pre-trained embedding model, allowing efficient similarity-based retrieval [28]. The vectorized representations were stored in a high-dimensional vector database, ensuring scalability and fast query processing.

### *B. ADVANCED CONTEXT RETRIEVAL STRATEGIES*

To enhance the quality and relevance of summaries generated by LLMs, advanced retrieval strategies were implemented beyond basic similarity-based retrieval [29].

These strategies aim to improve contextual precision and coverage by intelligently structuring and expanding queries. Two such techniques-Hierarchical Chunk Retrieval and Query Expansion with RAG Fusion [30] are explored, each designed to capture a broader semantic range and preserve critical contextual information for downstream summarization.

#### 1) HIERARCHICAL CHUNK RETRIEVAL

In this method, to improve retrieval precision, a hierarchical retrieval mechanism was adopted. The review corpus was initially divided into large parent chunks, preserving high-level context. Each parent chunk was further segmented into child chunks, which were independently embedded into the vector database. This hierarchical structure allowed fine-grained retrieval while maintaining broader contextual coherence [31]. When a user submits a query, the system retrieves child chunks most relevant to the query. The effectiveness of this approach depended on the granularity of segmentation—too few child chunks would make retrieval similar to traditional RAG, while too many could lead to fragmented context. The retrieved child chunks were then used as input for LLM-based summarization. This improves contextual coherence; as standard RAG often loses semantic flow when operating on isolated chunks. Retrieving entire parent documents ensures that the generation process retains topical alignment.

#### 2) QUERY EXPANSION WITH MULTI-PERSPECTIVE RAG FUSION

In this method, instead of relying on a single query representation, a multi-perspective retrieval approach was employed. Given a user query, an LLM-generated multiple variations of the query, each capturing a different aspect of the request. For each expanded query, the most relevant document chunks were retrieved from the vector database. To aggregate results from multiple queries, Reciprocal Rank Fusion (RRF) as shown below was applied,

Algorithm for Reciprocal Rank Fusion (RRF):
Input: Set of re-ranked list of document lists R = D1, D2, D3, ...Dn, where each Di is an ordered list of retrieved documents and a fusion parameter ‘k’.
Output: A globally re-ranked list of documents based on RRF scores.

Step 1: Initialize a dictionary ‘S’ for scores such that S$\leftarrow \phi$.
Step 2: For each ranked list Di R:
For each document d at rank r in Di
i) convert d to a string, ‘dstr’.
ii) if dstr $\notin$ S then, set S[dstr]$\leftarrow$0.
iii) Update the score:
S[dstr]$\leftarrow$S[dstr]] + (1/(r+k)).
Step 3: Sort documents in descending order of S[d].
Step 4: Return ranked list ‘Dfused’ with scores.

ensuring that frequently retrieved and highly ranked documents were prioritized [32]. This approach mitigated single-query biases and enhanced the diversity of retrieved information. The reranked documents were then used to generate an initial summary.

### *C. ITERATIVE REFINEMENT VIA MULTI-HOP RETRIEVAL*

While above retrieval methods offer valuable context, they may not fully capture the depth and completeness required for insightful summarization. To address this, an iterative, multi-hop refinement mechanism was introduced. This approach incrementally enriches the context using dynamic query reformulation and retrieval, ensuring that the final summary is both comprehensive and coherent. The following sections detail the components and mechanisms used in this multi-stage refinement process.

#### 1) ADAPTIVE INFORMATION ENRICHMENT WITH DSPY INTEGRATION

To ensure completeness and contextual depth, a multi-hop retrieval strategy was integrated using the Declarative Self-improving Python (DSPy) framework [33]. In this iterative process, an initial summary was generated from the retrieved context by each of the above-described methods. At each hop, an additional query was dynamically formulated using chain of thought, based on the existing summary, retrieving supplementary information from the vector database. Newly retrieved content was merged into the evolving summary, ensuring redundancy was minimized while preserving information gain. This iterative retrieval continued until a predefined convergence criterion or hop limit was reached.

#### 2) FINAL SUMMARIZATION AND INSIGHT EXTRACTION

After the retrieval-refinement cycle, the final summary was generated using the LLM, ensuring that all relevant aspects of the query were addressed. The refined summary provided a comprehensive synthesis of customer feedback, allowing business stakeholders to extract actionable insights without manually analyzing large volumes of text.

## IV. RESULTS AND ANALYSIS

This section presents a structured evaluation of the proposed system in three stages. First, we benchmark two non-LLM baselines—BM25 [34], [35] for sparse lexical retrieval and a fine-tuned BERT [36], [37] model for extractive response generation. Next, we perform component-level analysis to

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QnA Pair 1:
Context: I was billed twice for Gold membership. No refund has been issued despite repeated contact attempts. I hope this gets resolved soon.
Question: What complaint does the user have?
Answers: [{'text': 'No refund has been issued despite repeated contact attempts', 'answer_start': 40}]
Is Impossible: False

QnA Pair 2:
Context: I was billed twice for Gold membership. No refund has been issued despite repeated contact attempts. I hope this gets resolved soon.
Question: What service is being criticized?
Answers: [{'text': 'Gold membership', 'answer_start': 23}]
Is Impossible: False

QnA Pair 3:
Context: I was billed twice for Gold membership. No refund has been issued despite repeated contact attempts. I hope this gets resolved soon.
Question: What specific failure did the user face?
Answers: []
Is Impossible: True
```

**FIGURE 2. Response of Fine-tuned BERT model.**

evaluate the impact of different retrieval strategies: traditional RAG, hierarchical chunk retrieval, and RAG Fusion, each assessed without DSPy. Finally, we report the performance of the full pipeline, integrating hierarchical and fusion-based retrieval with the DSPy framework for iterative refinement. All models are evaluated using BERT Score, ROUGE, and RAGAS [38] metrics to assess semantic relevance, lexical coverage, and factual consistency, respectively.

The fine-tuned BERT model used in this evaluation is trained on the SQuAD dataset and operates in an extractive question answering setting. It selects the most relevant span from a provided context paragraph without generating new content. However, its performance is highly sensitive to input length and phrasing. BERT can only process a limited window of text, and when presented with lengthy customer reviews, it often fails to locate the correct answer or returns incomplete outputs. As shown in Fig. 2, the model relies on shallow spans of context and is unable to reason across multiple reviews or adapt to alternate phrasings of the same query. If the answer lies outside the attended span or is implied across sentences, the model either marks the answer as impossible or outputs irrelevant text. These limitations make it impractical for real-world QA tasks that involve long-form, noisy, and multi-turn user feedback.

Compared to the fine-tuned BERT model, BM25 performed more reliably for response generation in our evaluation. As a sparse lexical retrieval method, BM25 ranks review texts based on keyword overlap and term frequency, allowing it to identify relevant passages even in longer customer feedback. Unlike BERT, it is not limited by input length and can efficiently search across the entire corpus. As shown in Table 1, BM25 achieved higher scores than the BERT-based baseline across all metrics, though it still lagged behind LLM-based methods in semantic relevance and factual grounding. The F1 Score of BM25 peaked at 0.78, with an average precision of 0.77 and recall of 0.79, indicating limited semantic alignment with reference answers. Furthermore, the ROUGE scores were comparatively poor, with an average ROUGE-1 of 0.034, ROUGE-2 of 0.0352, and ROUGE-L of 0.032, reflecting minimal lexical overlap and summarization coherence.

**TABLE 1. BERT score and ROUGE score analysis for BM25.**

| Sl no. | BERT SCORE | | |
|---|---|---|---|
| | Precision | Recall | F1 Score |
| Average | 0.7665 | 0.7886 | 0.7768 |
| | ROUGE SCORE | | |
| | ROUGE-1 | ROUGE-2 | ROUGE-L |
| Average | 0.0347 | 0.0352 | 0.0327 |

These results highlight BM25's limitations in capturing semantic context, as it relies purely on sparse keyword matching. While BM25 is efficient and interpretable, its inability to model semantic nuances or handle paraphrased queries limits its suitability for feedback-driven natural language response generation.

On the other hand, Traditional Retrieval-Augmented Generation (RAG) improves language models by retrieving external data without retraining. Traditional RAG works by injecting the chunks that are retrieved using semantic similarity into the model's prompt after chunking the documents into embeddings and storing them in a vector database. This makes it possible to use current data to provide dynamic, domain-specific replies. The main drawback of traditional RAG is that it frequently recovers semantically related fragments while omitting more complex, interrelated meanings. Table 2 shows the performance of all the three LLM's for traditional RAG.

Even while the results might have a high similarity score, they might only cover a tiny percentage of pertinent information and lack thorough context. This limits the model's ability to truly comprehend complex queries and results in a large loss of information up to 90% of potentially helpful content may be missed.

The following table, Table 3 presents the evaluation of all the three LLM's, Meta-Llama-3-8B-Instruct-Lite, GPT-3.5-Turbo, and Gemini-1.5-Pro for both Hierarchical Chunk Retrieval and RAG Fusion without the integration

**TABLE 2.** ROUGE score, BERT score and RAGAS score analysis for traditional RAG.

| LLM | ROUGE SCORE | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Meta-Llama-3-8B-Instruct-Lite | | | gpt-3.5-turbo | | | gemini-1.5-pro | | |
| | ROUGE-1 | ROUGE-2 | ROUGE-L | ROUGE-1 | ROUGE-2 | ROUGE-L | ROUGE-1 | ROUGE-2 | ROUGE-L |
| TRADITIONAL RAG | 0.22 | 0.08 | 0.20 | 0.28 | 0.11 | 0.26 | 0.33 | 0.14 | 0.32 |
| | BERT SCORE | | | | | | | | |
| | Precision | Recall | F1 | Precision | Recall | F1 | Precision | Recall | F1 |
| | 0.71 | 0.72 | 0.71 | 0.76 | 0.78 | 0.77 | 0.81 | 0.82 | 0.82 |
| | RAGAS SCORE | | | | | | | | |
| | Context Recall | Factual Correctness | Faithfulness | Context Recall | Factual Correctness | Faithfulness | Context Recall | Factual Correctness | Faithfulness |
| | 1 | 0.59 | 0.56 | 1 | 0.67 | 0.65 | 1 | 0.76 | 0.74 |

**TABLE 3.** ROUGE score, BERT score and RAGAS score analysis (Without DsPy).

| LLM | ROUGE SCORE (Without DSPy) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Meta-Llama-3-8B-Instruct-Lite | | | gpt-3.5-turbo | | | gemini-1.5-pro | | |
| | ROUGE-1 | ROUGE-2 | ROUGE-L | ROUGE-1 | ROUGE-2 | ROUGE-L | ROUGE-1 | ROUGE-2 | ROUGE-L |
| Hierarchical Chunk Retrieval | 0.23 | 0.08 | 0.21 | 0.29 | 0.11 | 0.27 | 0.34 | 0.14 | 0.33 |
| RAG Fusion | 0.26 | 0.10 | 0.23 | 0.31 | 0.12 | 0.29 | 0.37 | 0.16 | 0.35 |
| | BERT SCORE (Without DSPy) | | | | | | | | |
| | Precision | Recall | F1 | Precision | Recall | F1 | Precision | Recall | F1 |
| Hierarchical Chunk Retrieval | 0.77 | 0.75 | 0.76 | 0.79 | 0.80 | 0.79 | 0.83 | 0.84 | 0.84 |
| RAG Fusion | 0.79 | 0.78 | 0.78 | 0.83 | 0.82 | 0.82 | 0.86 | 0.87 | 0.86 |
| | RAGAS SCORE (Without DSPy) | | | | | | | | |
| | Context Recall | Factual Correctness | Faithfulness | Context Recall | Factual Correctness | Faithfulness | Context Recall | Factual Correctness | Faithfulness |
| Hierarchical Chunk Retrieval | 1 | 0.61 | 0.57 | 1 | 0.69 | 0.67 | 1 | 0.78 | 0.76 |
| RAG Fusion | 1 | 0.66 | 0.60 | 1 | 0.75 | 0.71 | 1 | 0.83 | 0.78 |

of DSPy. This baseline evaluation allows to contrast the strengths and limitations in the absence of dynamic and iterative reasoning introduced by DSPy. The BERT Score highlights the impact of retrieval structure on semantic similarity.

The analysis of our model relies on a set of test queries, as detailed in Table 4, which encompass customer feedback scenarios such as service satisfaction, billing issues, chatbot interactions, email support effectiveness, and the rationale behind one star app ratings. Moreover, Table 5 and Table 6,

**TABLE 4. Reference queries and ground truth.**

| Question | Ground Truth |
|---|---|
| Are customers happy with the service? | I am not happy with the service. My orders often have missing items, deliveries are delayed, and customer support is terrible. While I do appreciate the app's efficiency and the reliability of its delivery partners, these positives don't outweigh the high prices, poor customer service, and issues with specific restaurants. |
| What are issues with Billing? | I am really frustrated with the billing system. My orders often have missing items, and I don't get refunds for failed transactions. The app doesn't provide confirmation during payments, and there's a lack of options to report payment problems. On top of that, the pricing is unclear, there are billing discrepancies, and I even had to pay unfair cancellation fees. Finding the cancellation option is a hassle, and I can't order from multiple restaurants at once. Customer care doesn't help at all! |
| Are customers happy with chatbots? | I am not happy with the chatbots at all. They are unhelpful, frustrating to use, and don't resolve any of my issues. Their responses are so limited that I end up wasting time without getting a solution. |
| What are customers' opinions about email support? | I am completely dissatisfied with the email support. It feels like a scam because they don't even listen to us! The responses are either automated or take forever, and when they do reply, their assistance is totally ineffective. |
| Why did few customers give a one-star rating to the app? | I gave the app a one-star rating because the customer service is terrible, and they don't resolve technical issues. On top of that, they cancelled my order during a surge period, delivered incorrect orders with missing items, and provided no proper support options. It's so frustrating! |

also illustrate the expanded queries generated through the RAG Fusion process and Hierarchical Retrieval, highlighting how LLMs iteratively refine input queries to enhance retrieval precision.

Combining both dense and lexical signals through RAG Fusion and Hierarchical chunk retrieval, LLM's can access semantically aligned information. The results in Table 5 illustrate, how this refined pipeline enhances the quality of outputs by generating expanded queries.

Since the user queries are often too concise, it is efficient for each LLM to generate text for the reference queries. The retrieval is enhanced by allowing the LLM to reframe the query. These expanded queries assist in identifying relevant content via RAG Fusion. The LLM can use retrieved relevant information and generates a final response by using both original query and the retrieved context. This process improves the relevance and factuality in the generated responses.

Table 7 and Fig. 3 present the BERT Score components—Precision, Recall and F1—for the Hierarchical Chunk Retrieval and RAG Fusion strategies across three LLMs as specified in (1), (2), and (3)

$$Precision = \frac{\sum_{i=1}^{N}\sum_{j=1}^{M} w_{ij} s\left(x_i, y_j\right)}{\sum_{i=1}^{N}\sum_{j=1}^{M} w_{ij}} \tag{1}$$

$$Recall = \frac{\sum_{j=1}^{M}\sum_{i=1}^{N} w_{ij} s\left(x_i, y_j\right)}{\sum_{j=1}^{M}\sum_{i=1}^{N} w_{ij}} \tag{2}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{3}$$

The BERT Score table highlights the impact of retrieval structure on semantic similarity. RAG Fusion (without DSPy) as in Table 3 achieves the highest F1 score of 0.86, followed by Hierarchical Chunk Retrieval at 0.84, and traditional RAG at 0.82 as in Table 2. This shows that multi-query evidence aggregation in RAG Fusion captures broader semantic alignment, while Hierarchical Chunk Retrieval enhances cohesion but with slightly narrower scope.

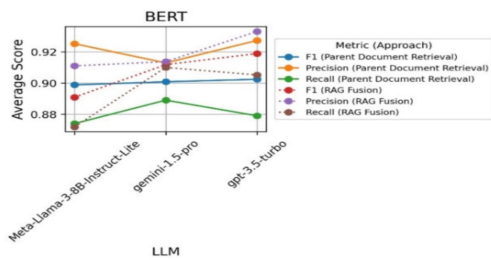


**FIGURE 3. BERT score comparison of LLMs with advanced RAG using DsPy.**

When DSPy is integrated, these values increase to 0.91 (RAG Fusion + DSPy) and 0.90 (Hierarchical Chunk + DSPy), confirming that reasoning refinement adds value. However, the key takeaway is that the bulk of semantic gains—from 0.82 to 0.91—are driven by retrieval enhancements alone.

The ROUGE scores—ROUGE-1, ROUGE-2, and ROUGE-L—are shown in Table 8 and Fig. 4 are computed as specified in (4) and (5),

$$ROUGE_N = \frac{\sum_{s \in S}\sum_{s \in S} Match\,(w)}{\sum_{s \in S}\sum_{s \in S} Count\,(w)} \tag{4}$$

$$ROUGE_L = \frac{LCS\,(X, Y)}{|Y|} \tag{5}$$

As shown, ROUGE results further illustrate how retrieval design affects lexical precision. RAG Fusion without DSPy

**TABLE 5. Queries generated by the LLMs for reference queries in RAG fusion.**

| Question ID | Question | Meta-Llama-3-8B-Instruct-Lite | gpt-3.5-turbo | gemini-1.5-pro |
|---|---|---|---|---|
| Q1 | Are customers happy with the service? | • "Are customers satisfied with the service?"<br>• "How do customers rate the service?"<br>• "What do customers<br>• think about the service?"<br>• "Are customers pleased with the service?" | • "Customer satisfaction metrics for service"<br>• "Service quality customer feedback analysis"<br>• "How to measure customer happiness with service?"<br>• "Surveys for assessing customer service satisfaction" | • " How satisfied are customers with the service provided?"<br>• " What are Customer reviews about the service?"<br>• "What are Common complaints about service?"<br>• "How does service measure Customer happiness?" |
| Q2 | What are issues with Billing? | • "What are the common issues with billing in healthcare?"<br>• "What are the most common billing errors in medical claims?"<br>• "What are the consequences of delayed or incorrect billing in healthcare?"<br>• "What are the best practices for resolving billing disputes in healthcare?" | • "Common billing errors and how to avoid them"<br>• "What are the most frequent customer complaints regarding billing?"<br>• "Best practices for transparent and accurate billing processes"<br>• "How to resolve billing disputes effectively?" | • "What are the typical billing errors, and how can they be prevented?"<br>• "What billing problems do customers raise the most in their complaints?"<br>• "What are the key strategies for maintaining clear and precise billing processes? "<br>• "How to resolve billing disputes effectively?" |
| Q3 | Are customers happy with chatbots? | • "Are customers satisfied with chatbot interactions?"<br>• "Do customers prefer human customer service over chatbots?"<br>• "How do customers rate their experience with chatbots?"<br>• "Are customers more likely to return to a business with good chatbot service?" | • "Customer satisfaction rates with chatbot interactions"<br>• "User experience with chatbots: positive vs. negative feedback"<br>• "Measuring chatbot effectiveness in improving customer happiness"<br>• "How do chatbots impact customer satisfaction compared to human agents?" | • "Customer satisfaction rates with chatbot interactions"<br>• "User experience with chatbots: positive vs. negative feedback"<br>• "Measuring chatbot effectiveness in improving customer happiness"<br>• "How do chatbots impact customer satisfaction compared to human agents?" |
| Q4 | What is customers' opinion about email support? | • "What are customers saying about email support?"<br>• "What are the opinions of customers on email support?"<br>• "What do customers think about email support?"<br>• "How do customers rate email support?" | • "Customer satisfaction with email support response time"<br>• "Common complaints regarding email support interactions"<br>• "Effectiveness of email support in resolving customer issues"<br>• "How customers perceive the tone and helpfulness of email support agents" | • "Customer satisfaction with email support response times"<br>• "Common complaints regarding email support interactions"<br>• "Effectiveness of email support in resolving customer issues"<br>• "How customers perceive the tone and helpfulness of email support agents" |
| Q5 | Why did few customers give one-star rating to the app? | • "Why did some customers give a one-star rating to the app?"<br>• "What are the reasons behind the one-star ratings given to the app?" | • "Reasons for 1-star app ratings"<br>• "Common complaints leading to 1-star app reviews"<br>• "Identifying causes of low app store ratings" | • "Reasons for 1-star app ratings"<br>• "Common complaints leading to 1-star app reviews"<br>• "User experience issues causing low app ratings" |

**TABLE 5.** ***(Continued.)*** **Queries generated by the LLMs for reference queries in RAG fusion.**

| | | | | |
|---|---|---|---|---|
| | | • "Why did a few customers rate the app with one star?"<br>• "What are the common issues that led to one-star ratings for the app?" | • "Analysing negative feedback from 1-star app reviews" | • "Analysing negative feedback from 1-star app reviews" |

records the highest ROUGE-1 at 0.37, followed by Hierarchical Chunk Retrieval at 0.34, as in Table 3 and traditional RAG at 0.33 as in Table 2. For ROUGE-L, the scores follow a similar pattern: 0.35 (RAG Fusion), 0.33 (Hierarchical Chunk), and 0.32 (RAG). These improvements indicate that RAG Fusion retrieves more phrasally diverse evidence, boosting overlap with references.

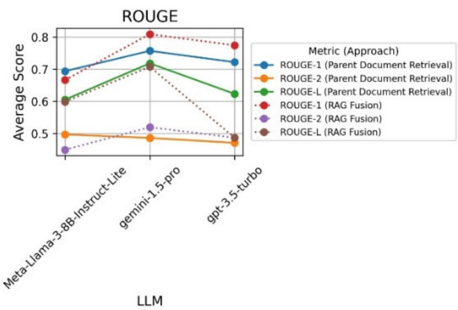


**FIGURE 4.** **ROUGE score comparison of LLMs with Advanced RAG using DsPy.**

When DSPy is added, ROUGE-1 scores rise to 0.80 (RAG Fusion + DSPy) and 0.75 (Hierarchical Chunk + DSPy), but the largest jump—from 0.33 to 0.80— already comes from retrieval quality. This reinforces that retrieval diversity has a strong influence on lexical fidelity.

Additionally, Table 9 and Fig. 5 report the RAGAS evaluation metrics: Context Recall, Factual Correctness, and Faithfulness. as specified in (6), (7) and (8)

$$\text{Context Recall} = \frac{|\text{Retrieved documents} \cap \text{Relevant documents}|}{|\text{Relevantdocuments}|} \quad (6)$$

$$\text{Faithfulness} = \frac{\text{factorial claims aligned with sources}}{\text{total factorial claims made}} \quad (7)$$

$$\text{Factual correctness} = \frac{\text{correct facts generated}}{\text{total facts generated}} \quad (8)$$

further illuminate these differences: RAGAS metrics reveal how retrieval affects factual alignment. All methods maintain 1.00 Context Recall, but differ in Faithfulness and Factual Correctness. RAG Fusion without DSPy leads with 0.78 Faithfulness and 0.83 Factual Correctness, while Hierarchical Chunk Retrieval scores 0.76 and 0.78, as in Table 3 respectively. Traditional RAG lags at 0.74 and 0.76, reflecting weaker grounding, as in Table 2.

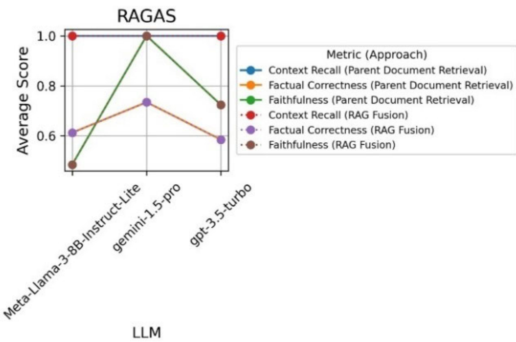


**FIGURE 5.** **RAGAS score comparison of LLMs with advanced RAG using DsPy.**

DSPy further improves these values to 1.00 Faithfulness (RAG Fusion + DSPy) and 1.00 (Hierarchical Chunk + DSPy), but the improvement from better retrieval alone (e.g., 0.78 to 1.00) is substantial. This shows that factual reliability depends primarily on how well the retrieval strategy captures meaningful and complete evidence.

In Table 7 for the RAG Fusion method—which leverages multiple query expansions and Reciprocal Rank Fusion to enhance retrieval diversity—a similar pattern emerges with nuanced differences. GPT-3.5-Turbo maintained robust performance, with a Precision of 0.9331 and an F1 of 0.9189, underscoring its adeptness at integrating augmented contextual information. Meta-Llama-3-8B-Instruct-Lite exhibited notable Recall (0.8740) and an F1 of 0.8909, albeit with slightly lower Precision. Gemini-1.5-Pro sustained high Recall (0.9100) and an F1 of 0.9118, reinforcing its reliability across retrieval strategies. In the Table 8 ROUGE evaluation under RAG Fusion, GPT-3.5-Turbo recorded consistent scores with ROUGE-1 at 0.7738 and ROUGE-L at 0.4865. Meta-Llama-3-8B-Instruct-Lite performed reliably, achieving a ROUGE-1 of 0.6934 and ROUGE-2 of 0.4965, while Gemini-1.5-Pro excelled once again with a ROUGE-1 of 0.8087 and ROUGE-L of 0.7178. In the Table 9, RAGAS metrics for RAG Fusion reveal that GPT-3.5-Turbo maintained perfect Context Recall (1.0000) with enhanced Faithfulness (0.7233), Meta-Llama-3-8B-Instruct-Lite showed stable Faithfulness (0.4843) and Factual Correctness (0.6120), and Gemini-1.5-Pro continued to excel with Faithfulness (1.0000) and Factual Correctness (0.7340).

**TABLE 6.** Generated text for reference queries by each LLM.

| LLM | | Generated Text | | |
|---|---|---|---|---|
| | | Meta-Llama-3-8B-Instruct-Lite | gpt-3.5-turbo | gemini-1.5-pro |
| Hierarchical Chunk Retrieval | Q1 | No, customers are not happy with the service. | No, customers are not happy with the service; they report numerous issues including missing items, delays, and poor customer support. | Customer sentiment is mixed. Some users appreciate the app's efficiency and delivery partners, while others express frustration with customer service, high prices, and restaurant-specific issues. |
| | Q2 | missing items in orders, refunds for failed orders, limited menu options, difficulty in finding the cancellation option, unrealistic expectations for online payments, inability to order from multiple restaurants at once, and lack of direct access to customer care. | The issues with billing include a lack of confirmation during the payment process, vague customer service responses, and insufficient options for reporting payment problems. | Inaccurate billing within the app leading to discrepancies between displayed and actual charges, lack of transparency in pricing, missing refunds for failed orders, and unfair cancellation fees |
| | Q3 | No, customers are not happy with chatbots. | No, customers are not happy with chatbots, as they find them unhelpful and frustrating for resolving issues. | No, customers are not happy with the chatbots. They find them unhelpful and frustrating due to their limited responses and inability to resolve issues effectively. |
| | Q4 | Customers are generally dissatisfied with the email support provided by the company, feeling that it is a scam and that they are not being heard. They are frustrated with the lack of response and the automated replies they receive | Customers find email support to be terrible, with long response times and ineffective assistance | They describe it as "terrible," "a scam," and express frustration with its slow response times and lack of effective resolution. |
| | Q5 | customers gave one-star ratings to the app because of bad customer service and technical issues. | Few customers gave one-star ratings to the app because of poor customer service, unresolved technical issues, fraudulent behavior, and a lack of support options. | Customers gave one-star ratings due to issues like order cancellations during surge periods, incorrect order fulfillment (missing items), and unsatisfactory customer service responses. |
| RAG Fusion | Q1 | No, customers are not happy with the service. | No, customers are largely unhappy with the service due to high charges, poor customer support, and frequent order issues. | Customer sentiment is mixed, with some users appreciating the app's efficiency and delivery partners, while others, are unhappy with the high charges, terrible customer support, and persistent order issues. |
| | Q2 | The issues with billing include inflated delivery charges, high delivery fees, and inflated menu prices, as well as a lack of transparency in pricing and a tendency for errors in payment processing and refunds. | Key billing issues include high delivery charges, fluctuating GST, lack of transparency, inaccurate billing discrepancies, and new charges like GST on packaging. | The billing system is frustrating, with high delivery charges, fluctuating GST, and a lack of transparency in cost calculations, compounded by discrepancies between displayed and actual charges, missing refunds for failed orders, and unfair cancellation fees, including GST on packaging. |
| | Q3 | No, customers are not happy with chatbots. | No, customers are generally unhappy with chatbots, finding them unhelpful and frustrating for resolving issues. | Customers are dissatisfied with the chatbots, describing them as unhelpful and frustrating because of their limited responses and inability to effectively resolve issues. |

**TABLE 6.** *(Continued.)* **Generated text for reference queries by each LLM.**

| | | | | |
|---|---|---|---|---|
| | **Q4** | Customers are generally dissatisfied with the email support provided by the company, feeling that it is a scam and that they are not being heard. They are frustrated with the lack of response and the automated replies they receive. | Customers find email support to be ineffective and frustrating, often receiving automated responses without resolution to their issues. | The email support is completely ineffective and frustrating, offering only automated replies that never resolve any issues, making it a waste of time for customers seeking assistance. |
| | **Q5** | The main reason why some customers gave the app one star is because of the poor service and experiences they had with the app, including issues with customer support, delivery, and pricing. | Customers gave one-star ratings due to poor customer service, high delivery charges, missing items in orders, and unhelpful automated support. | Customers frequently give one-star ratings due to terrible customer service, unresolved technical issues, order cancellations during surge periods, incorrect orders with missing items, and the lack of proper support options, leading to widespread frustration. |

**TABLE 7.** **BERT score for hierarchical Chunk retrieval and RAG fusion.**

| **LLM** | | **BERT SCORE** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | **Meta-Llama-3-8B-Instruct-Lite** | | | **gpt-3.5-turbo** | | | **gemini-1.5-pro** | | |
| | | **Precision** | **Recall** | **F1** | **Precision** | **Recall** | **F1** | **Precision** | **Recall** | **F1** |
| **Hierarchical Chunk Retrieval** | **Q1** | 0.9280 | 0.8556 | 0.8903 | 0.9253 | 0.8865 | 0.9055 | 0.9208 | 0.8952 | 0.9079 |
| | **Q2** | 0.9015 | 0.8754 | 0.8883 | 0.9146 | 0.8694 | 0.8914 | 0.8924 | 0.8692 | 0.8807 |
| | **Q3** | 0.9303 | 0.8656 | 0.8968 | 0.9303 | 0.8656 | 0.8968 | 0.9429 | 0.9169 | 0.9297 |
| | **Q4** | 0.9230 | 0.9001 | 0.9114 | 0.9230 | 0.9001 | 0.9114 | 0.8840 | 0.8698 | 0.8768 |
| | **Q5** | 0.9433 | 0.8735 | 0.9071 | 0.9433 | 0.8735 | 0.9071 | 0.9246 | 0.8941 | 0.9091 |
| | **Average** | 0.9252 | 0.8740 | 0.8988 | 0.9273 | 0.8790 | 0.9024 | 0.9129 | 0.8890 | 0.9008 |
| **RAG FUSION** | **Q1** | 0.9280 | 0.8556 | 0.8903 | 0.9225 | 0.8819 | 0.9017 | 0.9139 | 0.8991 | 0.9064 |
| | **Q2** | 0.8677 | 0.8590 | 0.8634 | 0.9263 | 0.8969 | 0.9113 | 0.9080 | 0.9129 | 0.9104 |
| | **Q3** | 0.9303 | 0.8656 | 0.8968 | 0.9246 | 0.9129 | 0.9187 | 0.9265 | 0.9245 | 0.9255 |
| | **Q4** | 0.9230 | 0.9001 | 0.9114 | 0.9494 | 0.9170 | 0.9329 | 0.9195 | 0.9151 | 0.9173 |
| | **Q5** | 0.9058 | 0.8793 | 0.8924 | 0.9428 | 0.9172 | 0.9298 | 0.9005 | 0.8984 | 0.8995 |
| | **Average** | 0.9110 | 0.8719 | 0.8909 | 0.9331 | 0.9052 | 0.9189 | 0.9137 | 0.9100 | 0.9118 |

This comparative analysis reveals that Gemini-1.5-Pro consistently outperforms its counterparts across Hierarchical Chunk Retrieval and RAG Fusion, demonstrating superior scores in BERT Score, ROUGE, and RAGAS metrics. GPT-3.5-Turbo maintains strong precision and balanced outputs, while Meta-Llama-3-8B-Instruct-Lite presents reliable yet improvable performance, particularly in Faithfulness and Factual Correctness. Notably, RAG Fusion enhances model performance, most prominently in Faithfulness and Factual Correctness, by leveraging enriched query contexts and dynamic retrieval strategies.

## V. FUTURE RESEARCH DIRECTIONS

The immediate focus of future work will be on improving the system's scalability and responsiveness in real-world settings. We plan to investigate model compression techniques and retrieval-side optimizations—such as caching and context filtering—to reduce inference latency and enable deployment on resource-constrained systems. We will also explore the use of distilled or quantized LLMs to retain performance while lowering computational cost. In the longer term, we aim to incorporate real-time adaptation through user-in-the-loop mechanisms. By leveraging implicit feedback—such

**TABLE 8.** ROUGE score for hierarchical Chunk retrieval and RAG fusion.

| LLM | | ROUGE SCORE | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Meta-Llama-3-8B-Instruct-Lite | | | gpt-3.5-turbo | | | gemini-1.5-pro | | |
| | | ROUGE-1 | ROUGE-2 | ROUGE-L | ROUGE-1 | ROUGE-2 | ROUGE-L | ROUGE-1 | ROUGE-2 | ROUGE-L |
| Hierarchical Chunk Retrieval | Q1 | 0.5833 | 0.4879 | 0.5167 | 0.7667 | 0.5786 | 0.6556 | 0.8563 | 0.5578 | 0.7044 |
| | Q2 | 0.7500 | 0.5614 | 0.6380 | 0.7500 | 0.5614 | 0.6380 | 0.6658 | 0.4571 | 0.5430 |
| | Q3 | 0.5722 | 0.4430 | 0.5278 | 0.7500 | 0.4255 | 0.6773 | 0.8580 | 0.4811 | 0.7310 |
| | Q4 | 0.8115 | 0.4816 | 0.6833 | 0.6519 | 0.3892 | 0.5764 | 0.5873 | 0.3851 | 0.8769 |
| | Q5 | 0.7500 | 0.5087 | 0.6577 | 0.7824 | 0.5167 | 0.6743 | 0.8158 | 0.5472 | 0.7336 |
| | Average | 0.6934 | 0.4965 | 0.6047 | 0.7214 | 0.4703 | 0.6228 | 0.7566 | 0.4857 | 0.7178 |
| RAG FUSION | Q1 | 0.5833 | 0.4879 | 0.5167 | 0.6880 | 0.4659 | 0.5754 | 0.8866 | 0.5750 | 0.7159 |
| | Q2 | 0.5932 | 0.4056 | 0.5932 | 0.7694 | 0.5167 | 0.7048 | 0.8500 | 0.5295 | 0.7750 |
| | Q3 | 0.5722 | 0.4430 | 0.5278 | 0.7048 | 0.4750 | 0.4750 | 0.7785 | 0.5500 | 0.6833 |
| | Q4 | 0.8115 | 0.4816 | 0.6833 | 0.7500 | 0.4430 | 0.6611 | 0.8685 | 0.5038 | 0.7574 |
| | Q5 | 0.7698 | 0.4259 | 0.6710 | 0.8412 | 0.5318 | 0.7360 | 0.6600 | 0.4369 | 0.6035 |
| | Average | 0.6660 | 0.4488 | 0.5984 | 0.7738 | 0.4865 | 0.4865 | 0.8087 | 0.5190 | 0.7070 |

**TABLE 9.** RAGAS score analysis for hierarchical Chunk retrieval and RAG fusion.

| LLM | | RAGAS SCORE | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Meta-Llama-3-8B-Instruct-Lite | | | gpt-3.5-turbo | | | gemini-1.5-pro | | |
| | | Context recall | Faithfulne-ss | Factual Correct-ness | Context recall | Faithfulness | Factual Correct-ness | Context recall | Faithfulne-ss | Factual Correct-ness |
| Hierarchical Chunk Retrieval | Q1 | 1.0000 | 0.3500 | 0.5700 | 1.0000 | 0.3500 | 0.5700 | 1.0000 | 1.0000 | 0.6700 |
| | Q2 | 1.0000 | 0.5714 | 0.4200 | 1.0000 | 0.6667 | 0.3100 | 1.0000 | 1.0000 | 0.6200 |
| | Q3 | 1.0000 | 0.3500 | 0.6400 | 1.0000 | 1.0000 | 0.6700 | 1.0000 | 1.0000 | 0.8300 |
| | Q4 | 1.0000 | 0.8000 | 0.8300 | 1.0000 | 1.0000 | 0.6700 | 1.0000 | 1.0000 | 0.8000 |
| | Q5 | 1.0000 | 0.3500 | 0.6000 | 1.0000 | 0.3500 | 0.6000 | 1.0000 | 1.0000 | 0.7500 |
| | Average | 1.0000 | 0.4843 | 0.6120 | 1.0000 | 0.7233 | 0.5840 | 1.0000 | 1.0000 | 0.7340 |
| RAG FUSION | Q1 | 1.0000 | 0.3500 | 0.5700 | 1.0000 | 0.3500 | 0.5700 | 1.0000 | 1.0000 | 0.6700 |
| | Q2 | 1.0000 | 0.5714 | 0.4200 | 1.0000 | 0.6667 | 0.3100 | 1.0000 | 1.0000 | 0.6200 |
| | Q3 | 1.0000 | 0.3500 | 0.6400 | 1.0000 | 1.0000 | 0.6700 | 1.0000 | 1.0000 | 0.8300 |
| | Q4 | 1.0000 | 0.8000 | 0.8300 | 1.0000 | 1.0000 | 0.6700 | 1.0000 | 1.0000 | 0.8000 |
| | Q5 | 1.0000 | 0.3500 | 0.6000 | 1.0000 | 0.3500 | 0.6000 | 1.0000 | 1.0000 | 0.7500 |
| | Average | 1.0000 | 0.4843 | 0.6120 | 1.0000 | 0.7233 | 0.5840 | 1.0000 | 1.0000 | 0.7340 |

as corrections, reformulated queries, or preference signals—the system can incrementally refine its retrieval and generation behaviour without full-scale retraining. Given the bias concerns raised in manual feedback analysis, we also plan to study retrieval strategies that promote evidence diversity and apply fairness-aware ranking to reduce representational bias in responses. These efforts will be supported by real-world evaluations measuring latency, consistency, and user satisfaction under dynamic operating conditions.

## VI. CONCLUSION

This paper presents an effective framework for retrieving and generating feedback-based responses using large language models. By introducing a Hierarchical Chunk retrieval method and integrating multi-query fusion and iterative reasoning, the system addresses key limitations of both traditional extractive baselines and generic RAG-based methods. Experimental results across three LLMs show consistent improvements in semantic accuracy, lexical precision, and grounding quality, with performance gains of 15% over non-LLM baselines. By reducing the reliance on manual feedback analysis and supporting aspect-specific, query-based interpretation of reviews, the system offers a scalable and modular solution for real-world applications. Additionally, the architecture allows for flexible deployment, with parallelizable retrieval and generation stages that can be optimized for latency and throughput in production settings. This demonstrates the practical value of our approach for automating customer feedback analysis at scale.

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